

Cognitive computing method based on decoding psychological emotional states

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ABSTRACT

The current artificial intelligence is constrained to passively executing human command control. It cannot perceive, learn, and guide itself. Furthermore, the majority of these systems are unable to comprehend the human psychological cognitive state or emotional intention actively. To enhance the universality and generalization ability of existing emotion recognition methods, a psychological emotion state decoding method combining singular spectrum analysis and entropy measure is designed based on an electroencephalography signal. This method is developed to empower the cognitive computing system with the ability to perceive emotions. The entropy measure is accelerated by the vector dissimilarity criterion. Finally, dynamic sample entropy model learning is used to realize the cognitive calculation of psychological and emotional state recognition. The results demonstrated that the computational efficiency of the proposed decoding algorithm was significantly improved. The average recognition accuracy of positive and negative emotional states reached 82.78 ± 16.22 . The best recognition accuracy of the proposed electroencephalography emotion recognition method was 84.64 %. The proposed electroencephalography emotion recognition method demonstrated an accuracy of 64.15 % in recognizing cross-individual positive, neutral, and negative emotional states. The proposed emotion recognition method shows better universality and generalization performance, and can effectively identify people's psychological and emotional states.

1. Introduction

The field of psychological and emotional state recognition is an interdisciplinary research area that employs cutting-edge technologies, including those from psychology, cognitive science, and computer science. It serves as the foundation for the development of emotional artificial intelligence and represents one of the key components in the construction of machine emotional intelligence (Liu et al., 2022; Movassagh et al., 2021; Usman & Abdullah, 2023). The challenges of sentiment analysis include data imbalance, linguistic diversity, and complexity of emotional expression. To address these challenges, scientists developed the Geyser-Inspired Algorithm. In addition, the Multi-objective Snow Ablation Optimization Algorithm and other more effective optimization strategies have been developed (Ghasemi et al., 2024; Pandya et al., 2024). In general, people's psychological and emotional states can be transmitted through a variety of signals. One category is the emotional signal that can be directly observed from people's external behavior and characteristics, such as facial expression,

voice intonation, body posture, etc. The second category encompasses those emotional signals that are not readily discernible through direct observation of an individual's external behaviors and characteristics. These include Electroencephalography (EEG), electromyographic signals, electrocardiogram signals, skin conductance, pulse, heartbeat, skin temperature, and breathing signals (Aryavalli & Kumar 2023). Nevertheless, the recognition of emotions through facial expressions, voice, and other external behaviors and features is susceptible to the influence of social and personal subjective factors. This can impede the accuracy of emotion recognition. However, emotional recognition methods based on physiological signals can to some extent mitigate the impact of environmental and social factors, enabling an objective representation of individuals' psychological and emotional experiences (K Liu et al., 2023). The current methods for recognizing psychological and emotional states based on physiological signals are constrained by the fact that the emotional information conveyed by physiological signals is subject to significant individual differences. Furthermore, the emotional response of different individuals to the same audio-visual stimulus is

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often disparate, with each individual exhibiting a unique emotional response (Lu et al., 2021). Such individual differences give rise to some issues, including the lack of universality of recognition methods, the dependence of recognition models on data samples, and the limitation of emotional data from the same individual. Furthermore, the lack of robust recognition results is a common occurrence in most existing research on psychological and emotional state recognition based on physiological signals (S Liu et al., 2023). To overcome the individual diversity of emotional information in physiological signals and enhance the generalization ability of the recognition model, a method for decoding EEG signals based on Singular Spectrum Analysis (SSA) and entropy measure is designed. A cognitive computing method based on the decoding of brain electrical information to identify cross-individual emotional changes is constructed. This method endows the cognitive computing system with emotional cognition abilities comparable to those of humans.

The novelty of this study is as follows: (1) The entropy feature extraction of EEG signals is optimized and improved by using SSA decomposition and reconstruction signal processing methods. An EEG decoding method is designed based on the combination of SSA and entropy measures. (2) A vector dissimilarity judgment criterion is constructed to pre-judge the dissimilar vectors. Furthermore, the efficiency of the most time-consuming vector distance calculation step of the entropy measurement algorithm is optimized. (3) The time-domain contour of EEG dynamic entropy measure is represented by the entropy measure feature vector to enhance the generalization ability of EEG emotional pattern recognition.

The contributions of this study are as follows: (1) A novel EEG decoding method based on the combination of SSA and entropy measure is proposed. This method achieves high recognition accuracy through the EEG entropy feature and effectively realizes the performance optimization of the EEG decoding task. (2) A fast calculation method of the entropy measure is proposed by using vector dissimilarity judgment criteria. The efficiency of the most time-consuming step of vector distance calculation of entropy measure algorithm is optimized. The accelerated calculation of entropy measure is realized. (3) By integrating entropy measurement with machine learning, it is possible to identify auditory attention objects with high accuracy from EEG signals.

The content is divided into four sections. The first section presents a review of the current state of application of cognitive computing and methods for emotion recognition. The second section investigates methods for decoding psychological emotional states based on cognitive computing. The third section involves performance validation of the proposed methods. The fourth section presents the conclusions.

2. Literature review

2.1. Cognitive computing

Cognitive computing research provides valuable insights into the development of neuromorphic artificial intelligence technology and methods. Enhancing existing artificial intelligence technologies using cognitive computing methods is a primary focus for many researchers. For instance, Mi et al. (Mi et al., 2022) developed a concept cognitive computing system to achieve dynamic classification learning of annotated data in a dynamic environment. The system utilized concept reduction space as a fundamental knowledge retention mechanism, integrating new information into old knowledge by simulating the human learning process. Compared to other methods, this system demonstrated strong performance in dynamic classification. Althobaiti et al. (Althobaiti et al., 2021) proposed a cognitive computing-based intrusion detection system for addressing security concerns in industrial information physical systems. They employed a feature selection technique based on binary bacterial foraging optimization to select the optimal feature subset. Integrated with a gate cycle meta-model, the system achieved intrusion detection in industrial information physical

environments. Through optimization algorithms, the hyper-parameters of the gate cycle meta-model were tuned to improve detection accuracy.

2.2. Application of cognitive computing

Qu et al. (Qu et al., 2020) developed a decentralized paradigm for data-driven cognitive computing based on federated learning and blockchain. This approach ensured privacy protection through federated learning and efficiently addressed the "data silo" problem. Blockchain provided incentive mechanisms, and completed decentralization and robustness against poisoning attacks. The effectiveness of this method has been validated. In the context of decision-making capabilities for drone landing target selection, Li et al. (Li et al., 2022) used cognitive computing to enhance the understanding of relationships between virtual models. Information descriptions were integrated across the components of the system using unified ontology and semantic modeling techniques. The resulting integrated simulation facilitated decision-making among component systems based on a high-level architecture. The inference of ontology models provided decision choices for the actual operation of component systems. To achieve effective integration of robotic automation and human cognition, Shufei Li et al. (Shufei et al., 2021) proposed a predictable information-based cognitive manufacturing paradigm. This paradigm established bidirectional empathy in the execution loop by comprehensively understanding human and robot scenarios. Through spatiotemporal cooperative prediction, it provided a basis for prospective planning. The system has been verified to support self-organizing learning and task assignment.

2.3. Emotion recognition based on EEG signal

Although emotions are a fundamental component of human daily life, research on the computational analysis of auditory emotions is not yet sufficiently deep. Consequently, many researchers have delved into this area. Zheng et al. (Zheng et al., 2022) researched the differential aspects of EEG data in the time and frequency domains. They employed a linear graph convolutional neural network to establish an EEG emotion recognition method based on dynamic differential entropy and brain connectivity features. By extracting dynamic differential entropy features that simultaneously reflect EEG frequency and time-domain characteristics, they computed Pearson correlation coefficients, phase-locking values, and transfer entropy. This led to the creation of a brain connectivity matrix, which fused information from multiple electrodes to achieve accurate classification of emotional states. Kim D. H. et al. (Kim et al., 2024) proposed an information-theoretic method for identifying epileptic patterns in short-time EEG recordings. This method was designed to decipher abnormal epileptic EEG, including seizure and intermittent EEG, from short-term EEG recordings. Additionally, the authors introduced multi-scale entropy analysis, which reflects the inherent dynamic complexity of EEG. In comparison to existing methods, this approach demonstrated superior efficacy in discerning epileptic EEG patterns.

2.4. Mental and emotional states based on brain signals

Hrisca-Eva OD et al. (Hrisca-Eva & Lazar, 2021) modeled EEG signal as an auto-regressive process and used linear discriminant analysis, quadratic discriminant analysis, Mahalanobis distance, support vector machine, and K-nearest neighbor classifier to classify the extracted features. This model has played a certain role in improving the movement and communication ability of patients with disabilities. Goshvarpour A et al. (Goshvarpour & Goshvarpour, 2022) proposed an intelligent schizophrenia detection system based on the entropy weighting of EEG electrodes. By using the entropy measurement of EEG data disclosed by RepOD and the one-way analysis of variance test, a method for scoring and selecting EEG channels for each attribute was proposed. By using this system, the accuracy of diagnosis was increased

to 100 %. To enhance spatial representation learning, Cui et al. (Cui et al., 2020) introduced a region-asymmetric convolutional neural network to do emotional recognition. They employed continuous one-dimensional convolutional layers in the time feature extractor for region feature extractors learning to capture regional information, and asymmetric feature extractors. The model achieved recognition accuracy exceeding 95 %, surpassing existing methods significantly.

In conclusion, the majority of current research on emotion recognition based on EEG signals employs feature extraction methods derived from time domain, frequency domain, and time-frequency domain features. However, the universality and generalizability of EEG emotion recognition methods and models remain limited. These feature extraction methods ignore the fact that the EEG signal is nonlinear. How to extract pattern features and EEG information related to mental state from EEG signals is an important problem. Compared with the previous methods, the SSA method decomposes each order of SSA components from EEG signals. Then, based on the EEG SSA component, the entropy measure method is used to extract the EEG entropy feature. By constructing a vector dissimilarity criterion, the dissimilarity vector can be pre-judged before the distance calculation begins.

3. Calculation method of psychological emotional states

The research begins by investigating the factors influencing the performance of EEG decoding algorithms based on entropy measures. A novel EEG signal decoding algorithm that combines SSA with entropy solutions is proposed. To optimize the efficiency of entropy calculation, a rapid computation method based on a vector with dissimilarity criteria for entropy measures is introduced.

3.1. Decoding emotional states based on EEG

The electrical activity generated by the brain's neural system during reactions produces corresponding EEG signals on the scalp's surface. By identifying the characteristic features of EEG, various conscious activities within the brain can be interpreted. In recent years, with the increasing application of machine learning methods in biomedicine, significant progress has been made in EEG decoding research. EEG decoding effectively separates task-related components from brain waves and extracts task-related brain activities, achieving the goal of EEG recognition (Cao et al., 2023). Fig. 1 illustrates the fundamental process of EEG decoding.

In Fig. 1, it is evident that the first step in EEG signal decoding involves acquiring and pre-processing EEG signals, followed by the analysis and feature extraction of EEG data, which in the decoding process are crucial. EEG is a typical non-stationary, nonlinear, and complex physiological signal with a multi-component structure. For various specific EEG decoding tasks, it is sometimes difficult to extract enough EEG information to realize decoding by extracting EEG entropy features directly from the EEG signal. To optimize the decoding performance of EEG signals, various signal processing methods, such as discrete wavelet transform, empirical mode decomposition, short-time Fourier transform, etc., are used to extract features directly from time series. The EEG decoding based on entropy measure employs a two-step process. First, it performs effective signal decomposition and reconstruction of time

series through SSA. Second, it performs subsequent entropy feature extraction of time series decomposition components. This approach provides more effective feature information for decoding research and yields better decoding performance.

SSA is a nonlinear time series analysis method, and its basic principle is based on time series principal component analysis (Ly & El-Sayegh, 2023). The method begins with a dynamic analysis of time series, after which it constructs the trajectory matrix of the time series using phase space reconstruction. Furthermore, it can effectively decompose the trajectory matrix into a series of important component structures through Singular Value Decomposition (SVD) and diagonalization averaging. EEG signal usually has the characteristics of a non-stationary, nonlinear, random signal, etc. The SSA method has obvious advantages for processing this type of signal. In the process of extracting EEG entropy features, it is possible to integrate the signal processing method of SSA decomposition and reconstruction to separate the effective component structure of EEG time series (that is, the SSA component). This is a convenient approach for the subsequent extraction of EEG entropy measure features. The SSA is introduced into EEG information entropy feature extraction and a novel EEG decoding algorithm is designed based on the combination of entropy measures and SSA. Fig. 2 shows a new approach to EEG decoding that integrates SSA and entropy measures.

The SSA method in time-series analysis consists of two complementary stages: decomposition and reconstruction. The SSA decomposition process does embedding and SVD. During SSA decomposition, a novel phase reconstruction algorithm is employed to map one-dimensional time series into a new trajectory matrix. Under the window length w , for a time series $t(n)$, $n = 1, 2, 3, \dots, N$, it is mapped to a multidimensional vector space through phase space reconstruction, obtaining its trajectory matrix X , as shown in Eq. (1).

$$X = \begin{bmatrix} t(1) & t(2) & \dots & t(L) \\ t(2) & t(3) & \dots & t(L+1) \\ \vdots & \vdots & \ddots & \vdots \\ t(w) & t(w+1) & \dots & t(N) \end{bmatrix} \quad (1)$$

In the SSA procedure, the trajectory matrix undergoes SVD. The trajectory matrix is decomposed into the sum of w elementary matrices, as shown in Eq. (2).

$$\begin{cases} X_j = \sqrt{\lambda_j} U_j V_j^T \\ X = \sum_{j=1}^w \sqrt{\lambda_j} U_j V_j^T = X_1 + X_2 + \dots + X_w \end{cases} \quad (2)$$

In Eq. (2), λ_j is the eigenvalues and U_j represent the eigenvectors within covariance matrix XX^T . V_j^T is defined as $V_j = X^T U_j / \sqrt{\lambda_j}$. The reconstruction phase of SSA consists of two parts: grouping and diagonal averaging. In the grouping stage, based on the results of SVD, singular values are grouped with their corresponding left and right singular vectors U_j and V_j . Subsequently, the grouped matrix X_{c_k} is obtained according to Eq. (2). Using the results of SVD, a sub-script set $\{1, 2, \dots, w\}$ is formed from the order of the decomposition. On this basis, the sub-script set is sub-divided into subsets to obtain v subsets $c_1, c_2, \dots, c_v$ ($v \leq w$). The final result is the composite matrix X_{c_k} , which is merged into a new result matrix $\tilde{X} = \sum_{k=1}^v X_{c_k}$.

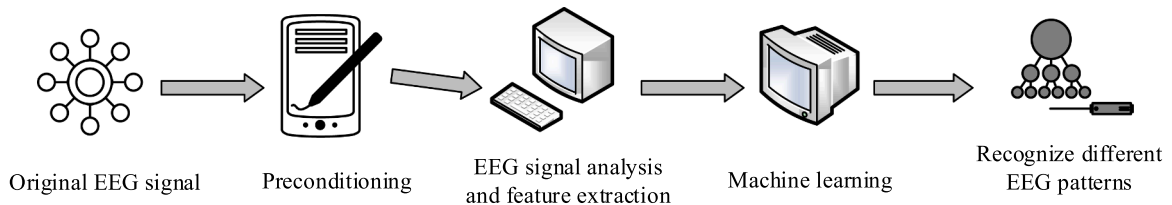


Fig. 1. Basic flow of emotional state decoding based on EEG signal.

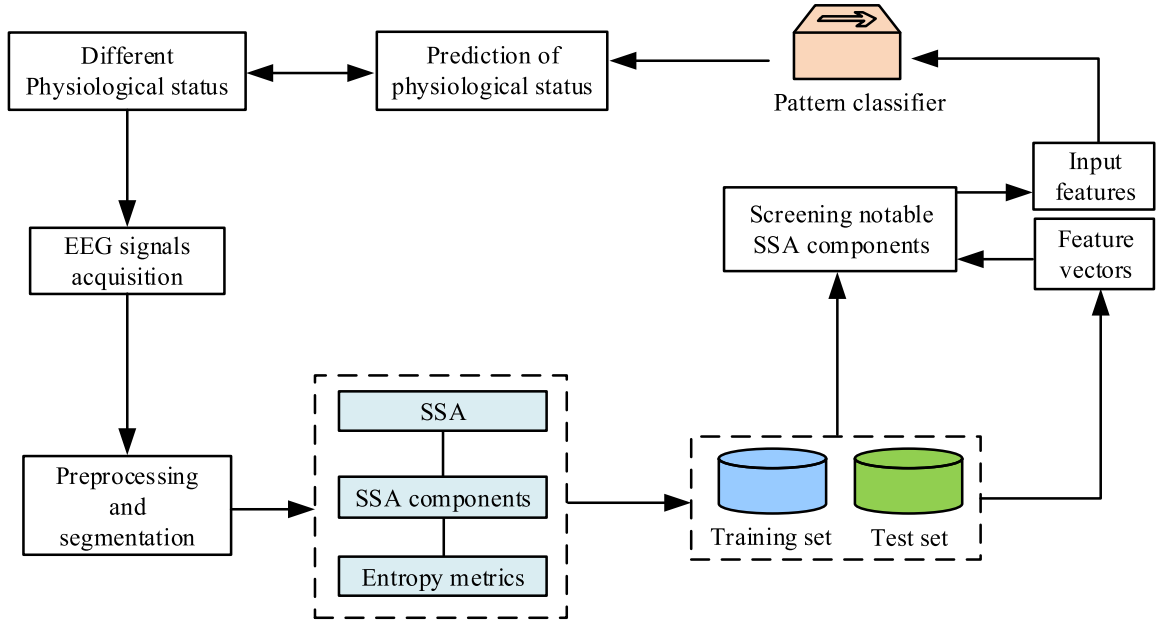


Fig. 2. Basic flow of EEG decoding based on SSA and entropy measure.

Given the $w \times L$ dimension of the obtained $\tilde{X}$, it can be rewritten as $\tilde{X} = (\tilde{x})_{ij}$, ($1 \leq i \leq w$, $1 \leq j \leq L$) based on its elements. The diagonal averaging in the SSA reconstruction process involves averaging all the diagonal elements $(\tilde{x})_{ij}$ on the diagonal $i + j = n + 1$ of the result matrix. This is reconstructed into $y(n)$, $n = 1, 2, 3, \dots, N$, mathematically defined as shown in Eq. (3).

$$y(n) = \begin{cases} \frac{1}{n} \sum_{i=1}^n \tilde{x}_{i,n-i+1}, & 1 \leq n \leq w \\ \frac{1}{w} \sum_{i=1}^w \tilde{x}_{i,n-i+1}, & w \leq n \leq L \\ \frac{1}{N-n+1} \sum_{i=n-L+1}^{N-n+1} \tilde{x}_{i,n-i+1}, & L+1 \leq n \leq N \end{cases} \quad (3)$$

The reconstructed time series signal $y(n)$ corresponds one-to-one with the generated result matrix. In the grouping process of SSA reconstruction, if $k = w$ holds, then $X = \tilde{X}$. In this case, the original time series signal $t(n)$ may be completely reconstructed from the generated

matrix X . The time series signal can be directly expressed as the sum of one-dimensional time series of each order, as shown in Eq. (4).

$$t(n) = \sum_{c_k=1}^w \sum_{n=1}^N \tilde{y}_{c_k}(n) \quad (4)$$

To better extract information about EEG entropy and optimize EEG decoding performance, the study integrates SSA steps into the process of extracting EEG entropy measurement features. Considering the EEG time series signal $s(n)$, $n = 1, 2, 3, \dots, N$, the time trajectory matrix of EEG is obtained through phase space reconstruction by varying the window length w . The eigenvalues and corresponding left and right singular vectors obtained through SVD are used to reconstruct various orders of SSA components. Finally, after SSA decomposition and reconstruction, the EEG signal $s(n)$ is represented by SSA components sum.

Fig. 3 illustrates the block diagram of brain entropy feature extraction based on the SSA components of EEG signals. The study employs the SSA algorithm, consisting of four steps: phase-space reconstruction of EEG, SVD, grouping, and diagonal averaging, to process EEG signals and obtain SSA components in each order. Since each SSA component is a

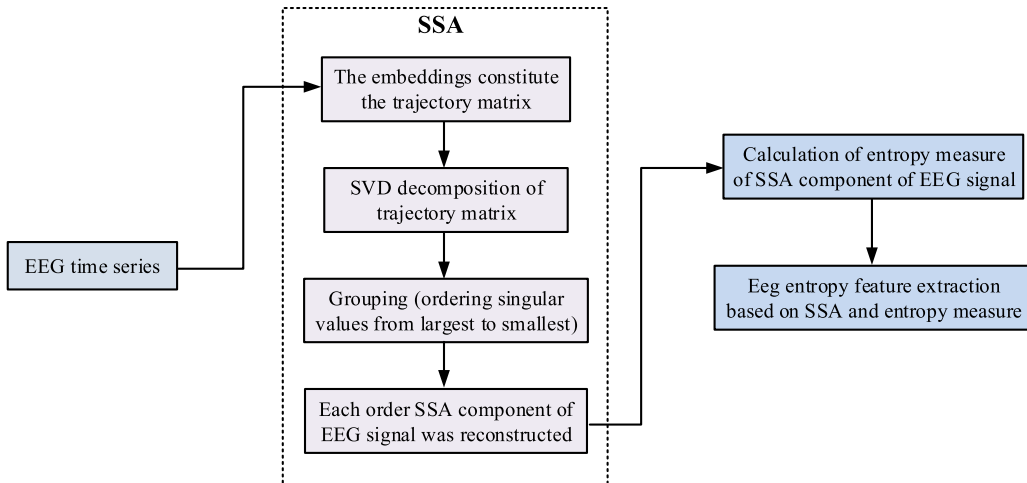


Fig. 3. EEG entropy feature extraction based on SSA components of EEG signals.

one-dimensional time series, the entropy measurement of these components provides information about the EEG signal. Four different entropy measurement methods, namely the Multi-scale, Sample, Fuzzy, and Approximate Entropy, are applied to analyze the EEG signals. Specifically, certain features are identified as "relevant features" in a specific feature vector space for a particular EEG task, while others are deemed "irrelevant features". The removal of irrelevant dimensions is a crucial step in EEG decoding, involving feature extraction or dimensionality reduction. Given this, a feature selection process is conducted on the SSA components of EEG through pairwise *t*-tests, where significance is set at $p = 0.05$. This identifies entropy features with significant differences, which are then utilized for EEG signal decoding.

3.2. Computational methods for emotion recognition cognition

EEG entropy measurement emerges as a significant nonlinear dynamical characteristic of EEG signals, gaining attention from scholars worldwide. However, commonly used entropy measurement methods such as Approximate Entropy, Sample Entropy, and Fuzzy Entropy face a notable challenge, i.e. high computational complexity (Garai et al., 2023). This complexity limits the real-time applicability of these entropy measures, especially in scenarios with stringent temporal requirements, such as real-time or online EEG signal decoding tasks.

A large amount of information will be lost if entropy measurement is used solely to determine criterion weights, because the interaction of decision information may be ignored (Gou et al., 2017; Gou et al., 2020). The fundamental objective of vector distance calculation is to perform statistical analysis on pattern vectors, ascertain whether they are analogous vectors in terms of distance between vectors, and conduct statistical analysis on the number of similar vectors (Olamat et al., 2022; Zhang et al., 2022). Currently, in entropy measurement calculations, distances between vectors are computed in phase space, followed by similarity assessments. However, vector distance calculations are time-consuming and do not necessarily require separate computations for different vectors. Therefore, the study proposes a vector with dissimilarity algorithm, introducing a criterion for determining vector dissimilarity directly. This aims to achieve rapid entropy computation including Sample, Approximate, and Multi-scale Entropy, bypassing the time-consuming calculation process.

Firstly, a decision quantity $Decision(i, j) \geq m \times r$ is constructed as a sufficient condition for determining the dissimilarity between the judgment vector $X_i^{(m)}$ and $X_j^{(m)}$. Its specific mathematical definition is given in Eq. (5).

$$Decision(i, j) = \left| \sum_{k=0}^{m-1} x_{i+k} - \sum_{k=0}^{m-1} x_{j+k} \right| \geq m \times r \quad (5)$$

In Eq. (5), m represents the embedded dimension of entropy measurement parameters. r represents the similarity tolerance of entropy measurement parameters. Regarding the calculation of approximate entropy and sample entropy, if the distance between any two m -dimensional vectors $X_i^{(m)}$ and $X_j^{(m)}$ is within the similarity tolerance r , they are considered similar, as shown in Eq. (6).

$$\begin{cases} |x_{i+k} - x_{j+k}| < r, \forall k, 0 \leq k \leq m-1 \\ |x_{i+k} - x_{j+k}| > r, \exists k, 0 \leq k \leq m-1 \end{cases} \quad (6)$$

When the decision quantity $Decision(i, j)$ satisfies $Decision(i, j) \geq m \times r$, the corresponding vectors $X_i^{(m)}$ and $X_j^{(m)}$ are necessarily dissimilar. If two m -dimensional vectors $X_i^{(m)}$ and $X_j^{(m)}$ are similar, then, by taking the absolute value of the difference of their corresponding m components, it will be smaller than $m \times r$, as shown in Eq. (7).

$$\max\{|x_{i+k} - x_{j+k}|\} < r, 0 \leq k \leq m-1 \Rightarrow \sum_{k=0}^{m-1} |x_{i+k} - x_{j+k}| < m \times r \quad (7)$$

Based on the mathematical definition of $Decision(i, j)$, given $Decision(i, j) \geq m \times r$, Eq. (8) can be derived.

$$\begin{cases} \left| \sum_{k=0}^{m-1} x_{i+k} - \sum_{k=0}^{m-1} x_{j+k} \right| \geq m \times r \\ \left| \sum_{k=0}^{m-1} x_{i+k} - \sum_{k=0}^{m-1} x_{j+k} \right| \geq m \times r \Leftrightarrow \left| \sum_{k=0}^{m-1} (x_{i+k} - x_{j+k}) \right| \geq m \times r \end{cases} \quad (8)$$

Furthermore, based on $|a| + |b| \geq |a + b|$, Eq. (9) can be obtained.

$$\sum_{k=0}^{m-1} |x_{i+k} - x_{j+k}| > m \times r \quad (9)$$

Eq. (9) represents the sum of the absolute values of the corresponding parts of any two m -dimensional vectors $X_i^{(m)}$ and $X_j^{(m)}$, obtaining a result greater than $m \times r$. Therefore, from Eq. (6) and Eq. (9), vector $X_i^{(m)}$ and $X_j^{(m)}$ are not similar, then Eq. (10) must be obtained.

$$\max\{|x_{i+k} - x_{j+k}|\} > r, 0 \leq k \leq m-1 \quad (10)$$

According to the Decision quantity $Decision(i, j) \geq m \times r$, it can be deduced that two vectors $X_i^{(m)}$ and $X_j^{(m)}$ are dissimilar. $Decision(i, j) \geq m \times r$ is a sufficient condition for two vectors $X_i^{(m)}$ and $X_j^{(m)}$ to be dissimilar. By using the vector with dissimilarity method, some vectors can be directly judged as dissimilar vectors in the calculation of Approximate entropy, Sample entropy, and Multi-scale entropy. In addition, the calculation of the distance between them and their time consumption can be completely omitted in the calculation process of the actual entropy measure. Additionally, it is worth noting that in the actual calculation process of entropy measure, the proportion of dissimilar vectors is in an obvious dominant position compared with similar vectors. Therefore, the vector with dissimilarity method can greatly optimize the efficiency of vector distance calculation in the process of entropy measure calculation to realize the acceleration of entropy measure calculation.

By adopting a time window with a width of t_w and moving along the time axis with a duration of Δt , from the time window, the EEG data sample entropy of is extracted out. The sample entropy expression, denoted as $DySampEn(m, r)$, is defined for embedded dimensions m and a similarity threshold r . Therefore, for a brainwave time series of length T , the dynamic sample entropy expression is given by Eq. (11).

$$DySampEn(m, r)_{(k)} = SampEn(m, r)_{(k)}, 1 \leq k \leq W \quad (11)$$

Fig. 4 illustrates the system diagram for emotion valence recognition. The study uses emotional images of different valence to stimulate the emotional states of participants while simultaneously collecting EEG signals related to emotions and cognition from the surface of the participant's scalp. Based on this, the EEG data undergo preprocessing and dimensionality reduction of features. Finally, using an interactive confirmation method, the emotion recognition algorithm and model are trained and tested. Emotion valence recognition is a method with strong generalization ability for cross-individual emotion valence recognition. Its purpose is to cross-validate different subject samples by using the emotional EEG of participants as test samples and the emotional EEG of other participants as training samples. Specifically, the study evaluates the overall performance of the emotion valence recognition algorithm by traversing each participant in the dataset and testing the emotion valence recognition accuracy of each participant.

Fig. 5 shows the structure of the selective attention decoding model based on a deep Long Short-Term Memory (LSTM) neural network. The proposed decoding model is composed of an attentional decoder network layer, a non-attentional decoder network layer, and an attentional classification network layer. The model is composed of an LSTM-Recurrent Neural Network (RNN), and its network depth is 6 levels, including the input layer. The Level 0 network layer is the input end of the model, corresponding to three input sample sequences: EEG

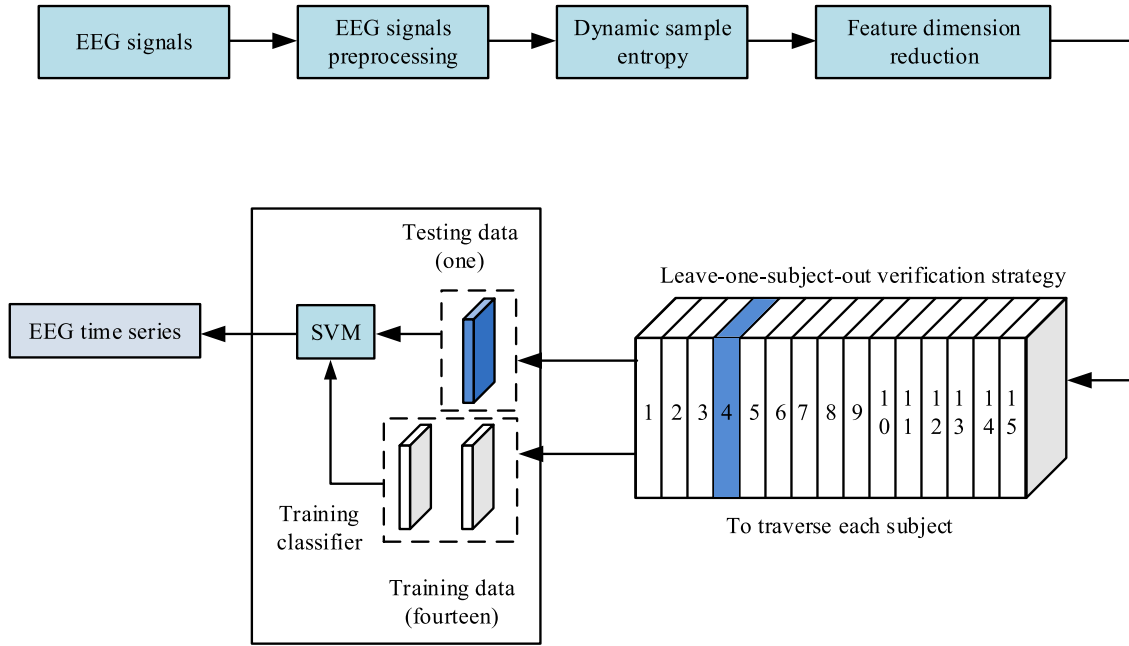


Fig. 4. Block diagram of emotion titers recognition system.

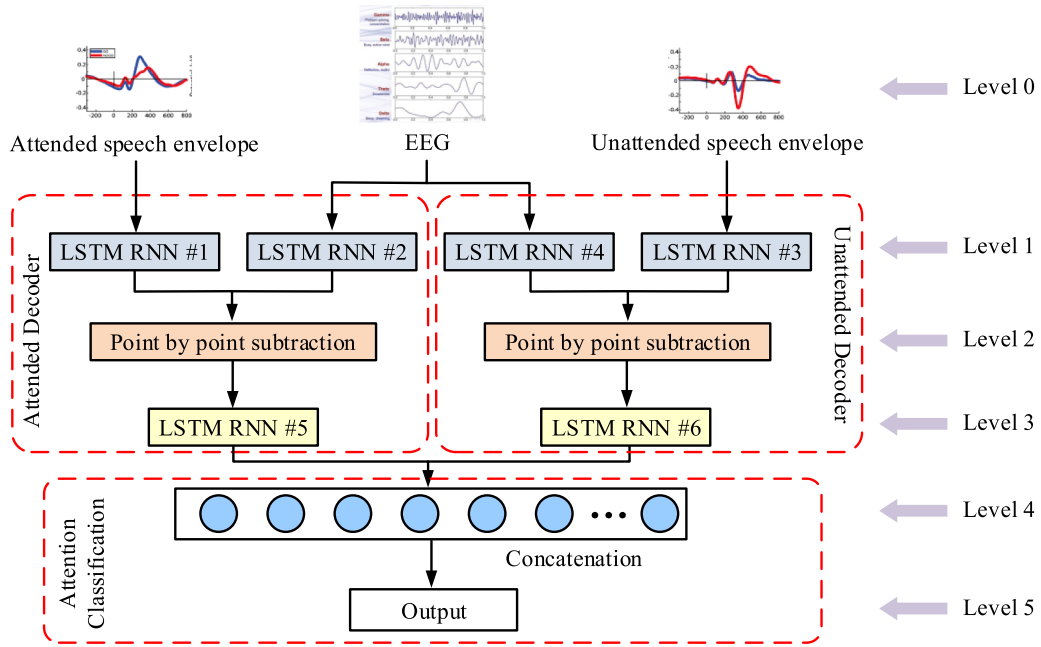


Fig. 5. Selective attention decoding model based on deep LSTM neural network.

sequence, attentional speech time-domain envelope sequence, and non-attentional speech time-domain envelope sequence. The Levels 1, 2, and 3 network layers in the deep network model are composed of attentional decoder and non-attentional decoder module networks in parallel. Both networks are three-level network layer structures stacked by three LSTM-RNN networks and a fusion layer. In the attention-decoder module network structure, the EEG sequence and attention-speech time-domain envelope sequence are first processed by an LSTM-RNN network layer. Then the data layer is fused by point-by-point subtraction. The non-attentional decoder module network structure is the same. The Levels 4 and 5 network layers in the deep network model are attention classification module networks, which are composed of 2 fully connected neural network layers. The Level 4 network layer is constituted

by the output layer of the attention decoder and the non-attention decoder module network in series, functioning as the input layer of the attention classification module network. The final Level 5 network layer is the output layer, which is composed of two neural units. The decoding model is a binary decoding task.

4. Analysis of experimental results

The experiment involved the performance evaluation of the proposed entropy measurement rapid calculation method using simulated signals and real EEG data. Cross-individual emotion state recognition was conducted on 12 participants using the Emotional EEG dataset of Shanghai Jiao Tong University, consisting of 6 males and 6 females

(average aged of 26.35 ± 3.64).

4.1. Algorithm performance validation

The study utilized video clips representing positive, negative, and neutral emotional states, with 5 clips for each emotion type (each lasting around 4 min). Participants' emotional states were observed, and EEG signals were collected using the ESI-128 EEG acquisition device manufactured by the American Neuroscan Company. EEG signals from 62 channels were sampled at a frequency of 1000 Hz, with electrode placement following the International Federation of Clinical Neurophysiology EEG electrode placement standards, as depicted in Fig. 6. In the database, each participant underwent three rounds of experiments, with at least a 7-day interval between each round. In each round, each person underwent 15 tests, resulting in a total of 750 EEG data samples representing different emotional states, with 225 samples for each emotion type. The EEG data used in the experiment used a time window of 4 s and the time of each window movement was 2 s. 29 sequential entropy features of EEG samples could be extracted from each channel EEG time series. The entropy measurement algorithm was implemented by MATLAB software. The experimental environment of all tests was Intel Core i7 CPU@3.2 GHz with 16GB RAM and Windows 7. The duration of the experimental EEG data collection was 60 s, the sampling rate was 500 Hz, and the length of EEG data samples was 30,000 points (60×500). The entropy measure had a data length of 30,000. The embedding dimension calculated by the entropy measure was 2 and the similarity tolerance was equal to 0.15.

Fig. 7 shows the analysis results of the computational resources required for the proposed algorithm. When the number of EEG data samples was 100, the Central Processing Unit (CPU) usage of the proposed method was 26.35 %. When the number of EEG data samples was 150, the CPU usage of the research method was 33.75 %. When the number of EEG data samples was 225, the CPU usage of the proposed method was 34.63 %. The CPU usage of the research method was always at a low level, and the system resource usage could meet the requirements of data production and access control.

For vector with dissimilarity algorithm test, the study compares it with the algorithm from reference (Wen et al., 2023). The average computation time of multi-scale entropy for EEG signals over different time intervals (1 to 8) is shown in Table 1. From Table 1, it is evident that the two algorithms exhibit significant differences in computation

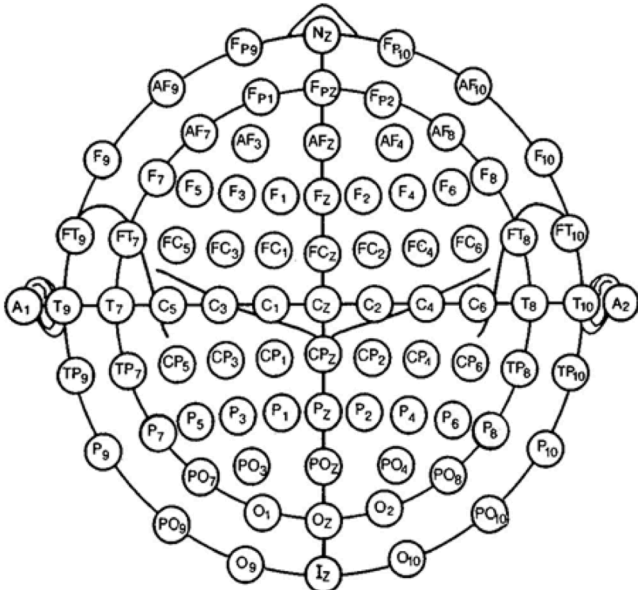


Fig. 6. Schematic diagram of placement of EEG electrodes.

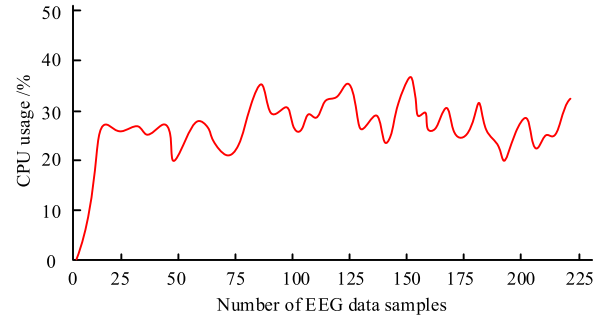


Fig. 7. Computing resources required by the algorithm.

time. Compared to the algorithm proposed in reference (Li et al., 2023), the proposed algorithm shows a maximum efficiency improvement of 48.3 % and an average improvement of 47.8 %.

The study processes EEG signals from two participants to understand the temporal distribution patterns of EEG entropy under different emotional valence conditions. Fig. 8 illustrates the time-domain profiles of EEG entropy features formed by dynamic sample entropy in 29 different time windows, representing the temporal distribution for two participants under positive and negative emotional conditions. In Fig. 8 (a), during the first experiment round, overall, entropy values under positive emotional states are slightly higher than those under negative emotional states. Fig. 8(b) indicates that, in the second and third experiment rounds, the entropy values of dynamic sample entropy profiles under positive emotional states are significantly higher than those under negative emotional states. In summary, based on the time-domain analysis of EEG dynamic sample entropy under positive and negative emotional states, it is found that the temporal variation characteristics of EEG entropy features contain information closely related to emotional valence-associated brain activity.

The study set up three different EEG samples for performance verification. In this experiment, subjects were asked to pay attention to the animal sounds played by the audio, and this experiment protocol was recorded as A. Subjects were asked to focus on listening to a continuous speech played by the audio, and this experimental protocol was denoted as B. The subjects were asked to keep their body and mind in a relaxed Rest state. The headphones did not play any audio signals. This experimental protocol was recorded as Rest. In this paper, the complex multi-scale entropy was extracted from 8-channel EEG signals using different parameters of similarity tolerance r . Table 2 shows that the composite multi-scale entropy with a similarity tolerance r equal to 0.1, 0.2, 0.3, and 0.4 was used as the input feature vector. In Table 2, under the composite multi-scale entropy with similarity tolerances r of 0.1, 0.2, 0.3, and 0.4, the average accuracy of Rest, A, and B recognition was 59.33 % ($p = 0.020$), 70.01 % ($p < 0.001$), 71.29 % ($p < 0.001$), and 55.52 % ($p = 0.068$), respectively. Therefore, the composite multi-scale entropy with a similarity tolerance r equal to 0.3 could obtain the best emotion recognition. The recognition accuracy of Rest, A, and B were 68.78 %, 77.84 %, and 67.24 %, respectively.

4.2. Analysis of recognition results

The study evaluated the performance of three rounds of EEG data in emotion valence recognition. Table 3 presents the results of positive and negative emotion recognition across individuals using a brainwave emotion pattern learning approach based on dynamic entropy. The results were expressed in terms of mean and standard deviation for 12 subjects. In the EEG emotion data corresponding to rounds 1, 2, and 3, the dynamic entropy-based brainwave emotion pattern learning achieved accuracy rates of 82.66 ± 13.25 , 82.86 ± 16.35 , and 82.81 ± 19.08 , with an average accuracy of 82.78 ± 16.22 . Throughout the test, the average recognition of positive and negative emotions accuracy was

Table 1
Time shift of EEG signal multi-scale entropy calculation average time.

Average time	Time interval							
	1	2	3	4	5	6	7	8
Reference (Wen et al., 2023)	1.223	1.846	2.3621	2.648	2.945	3.245	3.479	3.762
Vector with Dissimilarity	0.632	0.985	1.186	1.384	1.534	1.698	1.835	1.979
Time loss rate	0.483	0.466	0.498	0.477	0.479	0.477	0.473	0.474

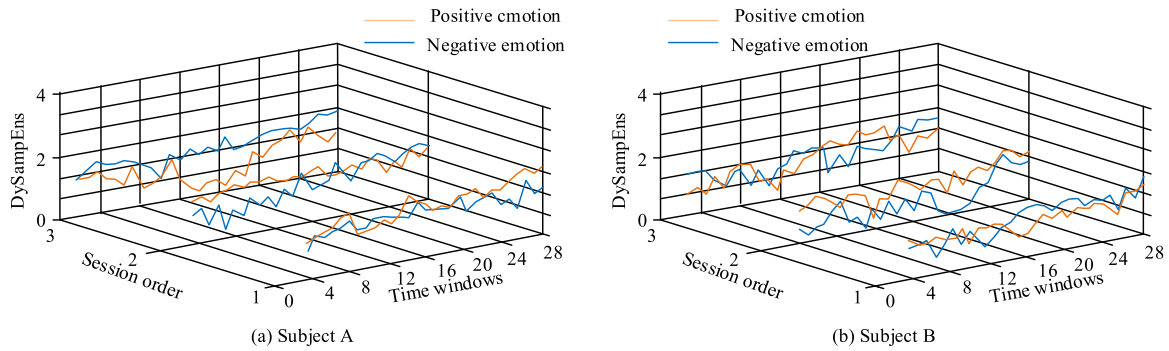


Fig. 8. Time-domain profile of the entropy of dynamic EEG samples of 2 subjects.

Table 2
Results of multi-scale entropy identification using different similarity tolerance parameters.

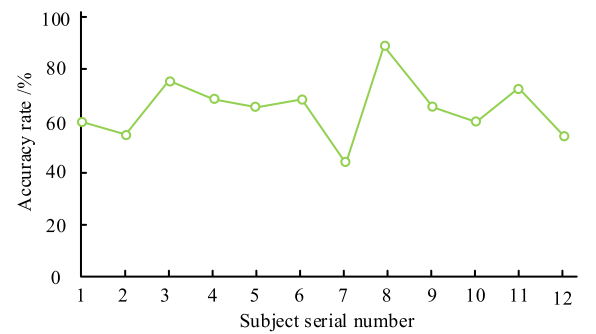
Recognition rate	Similarity tolerance r			
	0.1	0.2	0.3	0.4
Rest	63.25 %	69.85 %	68.78 %	55.17 %
A	61.76 %	77.18 %	77.84 %	62.23 %
B	52.97 %	63.01 %	67.24 %	49.17 %
Average recognition rate	59.33 %	70.01 %	71.29 %	55.52 %

Table 3
Results of emotion recognition.

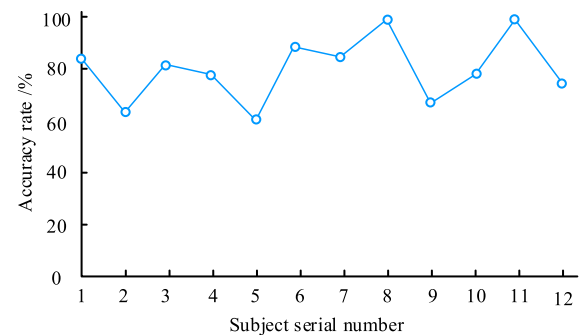
Experiment round	1	2	3	Mean value
Accuracy rate (%)	82.66 ±13.25	82.86 ±16.35	82.81 ±19.08	82.78 ±16.22
Sensitivity (%)	82.36 ±24.66	86.94 ±24.63	80.16 ±18.95	83.15 ±22.75
Specificity (%)	82.91 ±20.36	78.81 ±31.68	85.92 ±27.65	82.55 ±25.36
Error rate (%)	18.63 ±13.79	17.63 ±16.84	17.65 ±18.45	17.97 ±16.36
Sample size	150	150	150	450

quite similar. The experimental results demonstrated that the dynamic entropy-based brainwave emotion pattern learning could achieve emotion valence recognition across multiple individuals and exhibit strong robustness.

Fig. 9 illustrates the emotion recognition across individuals for the 12 subjects over the three rounds of experiments. As shown in Fig. 9(a), the accuracy rates for the 12 subjects were 86.67 %, 66.67 %, 83.33 %, 80.00 %, 63.33 %, 90.00 %, 86.67 %, 100 %, 70.00 %, 80.00 %, 100 %, and 76.67 %, respectively. The average accuracy was 84.64 %±11.38 %, surpassing the cross-individual emotion valence recognition performance in the aforementioned single-round experiments. This improvement was attributed to the general trend where the performance of learning algorithms tended to improve with an increase in the number of samples. Fig. 9(b) displayed the recognition results for positive, negative, and neutral emotions for the 12 subjects. In Fig. 9, the accuracy rates among the 12 subjects were 60.00 %, 55.56 %, 73.33 %, 66.67 %, 64.44 %, 66.67 %, 46.67 %, 84.4 %, 64.44 %, 60.00 %, 75.11 %, and 55.56 %, with an average accuracy of 60.98 %.



(a) Results of positive and negative emotion identification across individuals



(a) Results of positive, negative, and neutral emotion identification across individuals

Fig. 9. Results of emotion recognition based on dynamic entropy pattern learning.

64.44 %, 66.67 %, 46.67 %, 84.4 %, 64.44 %, 60.00 %, 75.11 %, and 55.56 %, with an average accuracy of 60.98 %.

The study utilized the Fourier transform, Classical Constant Q Transform (Classic-CQT), and Automatic Constant Q Transform (Auto-CQT) techniques to process EEG signals and obtain the percentage of the α frequency band. Regarding the Fourier transform, due to its large bandwidth for high frequencies, the total power carried by the bands was the sum of electrode spectral powers. In CQT, the band power was the average of electrode spectral powers for each band. Fig. 10(a)

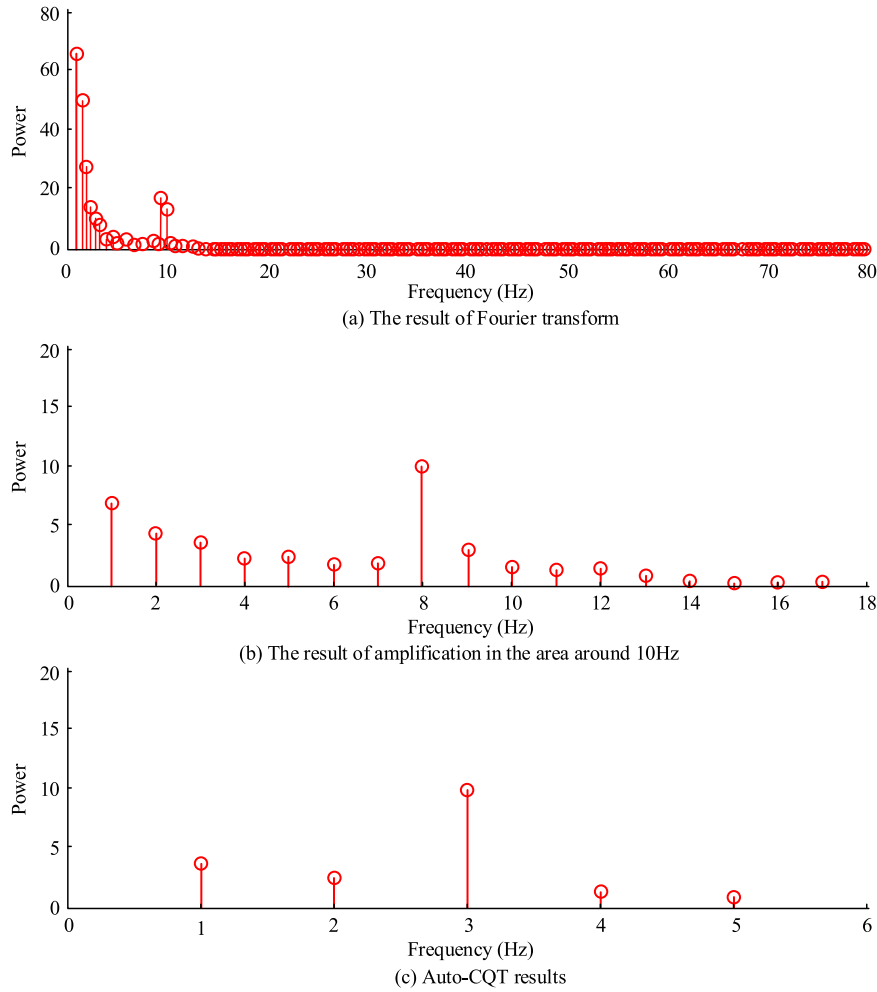


Fig. 10. Results of EEG spectrum analysis.

represented the results of the Fourier transform, showing a significant amplitude at 10 Hz, but adjacent spectral lines exhibited large amplitudes. Fig. 10(b) magnified the 10 Hz range of Fig. 10(a). Fig. 10(c) showed the results of Auto-CQT, indicating that CQT spectral lines exhibited better spectral features.

The study investigated the training and testing of an auditory selective attention encoding model based on LSTM deep networks using samples of nine different lengths. As depicted in Fig. 11(a), when the duration of sampled sequences did not exceed 1.5 s, the robustness of sound recognition accuracy for auditory attention targets across all participants was notably high. However, when the sampling duration exceeded 2 s, significant differences in the accuracy of target word recognition were observed among participants. The results indicated that, in sequences of varying lengths, the median accuracy for auditory attention targets was consistently close, as depicted in the box plot in Fig. 11(a). Fig. 11(b) illustrates the dependency between accuracy and the number of model training cycles. Under 5 training cycles, the recognition accuracy reached around 50 %, closely approximating chance level probability (50 %), suggesting insufficient training cycles and sub-optimal training. As the training cycles increased from 5 to 40, there was a significant improvement in recognition rates. With >45 training samples, the auditory selective attention decoding model based on LSTM deep networks achieved a prediction accuracy of around 95 %.

The research also involved partitioning brain electrodes to assess the performance impact of different brain regions' EEG information features on emotion valence recognition, as shown in Fig. 12(a). The partitions included the left frontal, right frontal, left posterior, and right posterior

areas, as well as anterior, posterior, left, and right hemispheres. To evaluate the performance impact of EEG information features from different brain regions on emotion valence recognition, the experimental data analyzed the recognition performance of positive and negative emotions across individuals under eight different brain region partitions. The results are depicted in Fig. 12(b). This study involved 15 participants, divided into 8 brain regions using EEG data. The accuracy of emotion valence recognition based on dynamic entropy-based EEG emotion map learning algorithm was found to be 84.36 %, 70.65 %, 78.44 %, 80.12 %, 70.08 %, 69.94 %, 80.23 %, and 74.63 %, respectively. This data corresponded to the frontal hemisphere, posterior hemisphere, left hemisphere, right hemisphere, left posterior region, right posterior region, left frontal lobe, and right frontal lobe regions of the brain. Despite having only 33 EEG electrodes, the anterior hemisphere exhibited the highest accuracy (84.36 %) in emotion valence judgment. Therefore, the selection of brain region locations and electrode channels was crucial for emotion valence judgment. The EEG activity in the anterior hemisphere of the brain might contain information most closely related to emotional valence, making it an ideal brain region for emotion valence judgment.

To compare with other emotion recognition systems, the study tested different algorithmic models on datasets. At the same time, other algorithms were reproduced for comparison, including Discrete Wavelet Transform (DWT), Empirical Mode Decomposition (EMD), and Short-time Fourier Transform (SFT). The test results of each subject are shown in Fig. 13. The data were divided into 10 equal fractions, and the ten-fold cross-verification of the data of a single subject was performed.

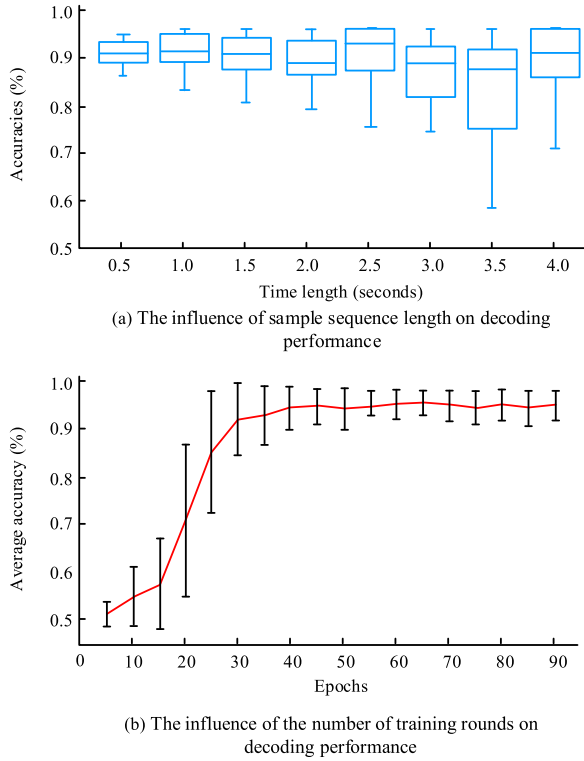


Fig. 11. Decoding performance analysis.

The results showed that the average accuracies of the proposed method, DWT, EMD, and SFT were 92.19 %, 85.51 %, 83.75 %, and 86.27 %, respectively. Compared to other algorithms, the emotion recognition system of the proposed method improved the recognition accuracy by >5 %. Compared with the baseline method, this method could obtain better performance of cross-individual emotion tier recognition. The main reason is that the input feature vector in the dynamic entropy-based EEG emotional pattern learning proposed in this paper is the time-domain contour of the dynamic sample entropy successively

connected through the sliding time window. This reflects the time-domain dynamic characteristics of the EEG entropy characteristics and represents the temporal state characteristics of the emotion, thereby enabling more effective representation of the EEG activity information related to the emotion intensity. Therefore, the EEG emotional pattern learning method based on dynamic entropy is more effective in monitoring emotional states derived from EEG signals, thereby enhancing the accuracy of inter-individual emotional tier recognition.

Table 4 shows the comparison between the proposed method and the experimental results of existing relevant studies. From the comparison in Table 4, the recognition rate of Rest and A is superior to the other two methods in the 3 classification auditory target recognition tasks achieved by the proposed methods. Although B is not the most advanced, it is at the same level as the existing relevant research reports. The above results fully show that the EEG entropy measure can be used to extract brain activity information about auditory target attention from EEG signals. The machine learning method is used to construct the auditory attention classifier. It can learn the rules and patterns related to auditory target attention from the EEG entropy feature data to realize the emotion recognition based on the EEG signal. In comparison to the other two methods, the proposed model necessitates a reduced number of channels for EEG decoding data, utilizing only eight channels of EEG data. This enables the realization of auditory selective attention decoding with high precision. It is evident that for particular EEG decoding tasks, the reduction of EEG decoding signal requirements, such as the duration of EEG time series and the number of EEG channels, will enhance the value of the approach in practice. The biggest disadvantage of the existing auditory selective attention decoding methods, especially the stimulus reconstruction methods, is that the EEG time series required for auditory attention decoding is too long, which greatly limits the value of the practical application. Therefore, the proposed auditory attention decoding method represents a promising avenue for future research and development.

5. Conclusion

The research focused on the content of individual psychological and emotional state recognition, combined the time series decomposition and reconstruction method with the EEG entropy measure, and designed

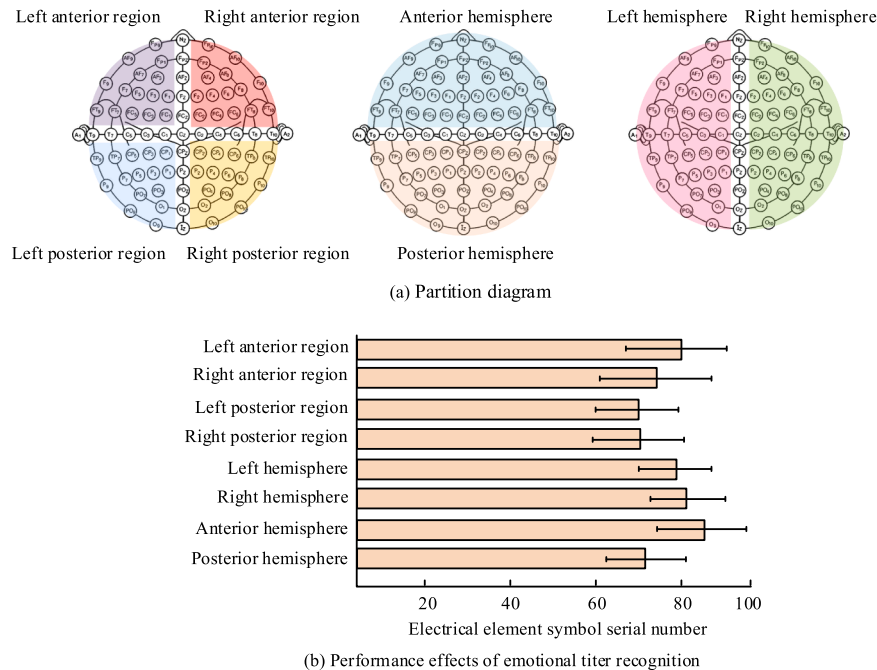


Fig. 12. Cross-individual accuracy of positive and negative emotion recognition in different brain regions.

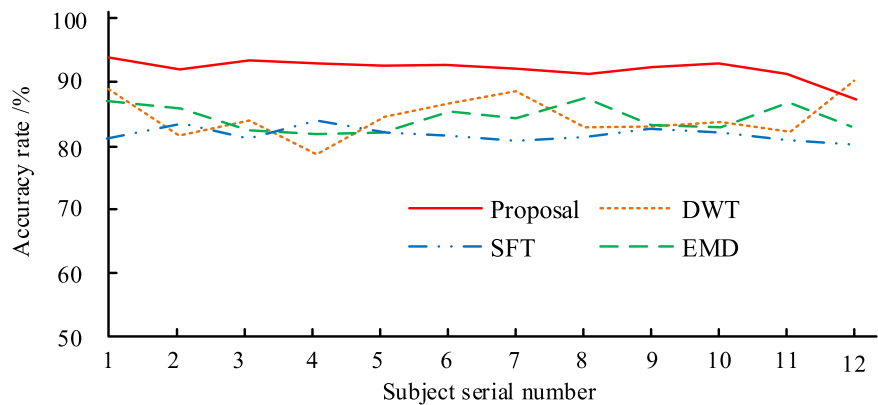


Fig. 13. Comparison of accuracy of 12 subjects under four algorithm tests.

Table 4
Feeder refinement analysis results.

Method	Task	Identification method	Passage	Result
Proposed method	Rest, A and B	Composite multi-scale entropy, Leave-one-out cross validation	8	70.36 %, 71.96 % and 62.74 %
Method of Li QL et al. (Li et al., 2023)	Attending left and attending right	Event-related potential templates, Leave-one-out cross-subject validation	32	67.32 %
Method of Kamble K S et al. (Kamble & Sengupta, 2023)	Direction of attention	Leave-one-out cross validation template matching approach	84	65.18 %

the EEG signal decoding method based on the combination of SSA and entropy measure. Given the time-consuming characteristics of the actual calculation process of entropy measure, a method was proposed to accelerate the calculation of entropy measure by using the vector dissimilarity criterion. The results showed that the EEG decoding method based on the combination of SSA and entropy measures could effectively optimize the performance of the EEG decoding task. Compared to the algorithm proposed in reference (Wen et al., 2023), the fast calculation method of entropy measures had an average increase of 47.8 % in computing efficiency. Finally, positive and negative emotion states were identified, and the best recognition accuracy of the proposed EEG emotion recognition method was 84.64 %. For cross-individual positive, neutral, and negative emotional states, the proposed EEG emotion recognition method achieved a recognition accuracy of 64.15 %. The average accuracy of emotion recognition was 82.78 ± 16.22 % in the EEG emotion data corresponding to the first, second, and third rounds. In the three rounds of experiments, the average accuracy of positive and negative emotion recognition was very similar, indicating that the EEG emotional pattern learning based on dynamic entropy can realize the emotion titer identification between multiple people, and has strong robustness. The average accuracy of the proposed method in the SEED data set was 92.19 %. Compared with DWT, EMD, and SFT algorithms, the proposed emotion recognition system improved the recognition accuracy by >5 %. The EEG decoding method proposed in this study had obvious advantages in processing EEG signals with non-stationary, non-linear, and multi-component structure characteristics. This method had better cross-individual emotion recognition performance, showing better universality and generalization ability. The constructed cognitive computing method for cross-individual emotional state recognition realizes the optimization and innovation of EEG emotional pattern recognition. It can effectively predict people's psychological and emotional states from EEG electrical signals. The

comparison with references (Li et al., 2023) and (Kamble & Sengupta, 2023) shows that the proposed method can learn emotional features related to EEG signals from EEG entropy feature data. In the process of this study, EEG signals may be subject to a variety of disturbances in the collection, such as EMG artifacts. These disturbances have the potential to affect the quality of the EEG signals, which in turn may reduce the accuracy of the emotion recognition process. In the future, the adaptive filter will be capable of automatically adjusting its parameters in accordance with the statistical characteristics of the EEG signal and noise, thereby enabling it to adapt to changes in the signal and noise. Furthermore, the filter parameters can be adjusted at the present time by utilizing the filter parameters obtained at a previous time point, thus achieving an optimal filter and enhancing the signal-to-noise ratio of the EEG output.

Abbreviation List

Full name	Abbreviation
Electroencephalography	EEG
Singular Spectrum Analysis	SSA
Long Short-Term Memory	LSTM
Recurrent Neural Network	RNN
Classical Constant Q Transform	Classic-CQT
Automatic Constant Q Transform	Auto-CQT
Empirical Mode Decomposition	EMD
Discrete Wavelet Transform	DWT
Short-time Fourier Transform	SFT
Singular Value Decomposition	SVD
Central Processing Unit	CPU

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CRedit authorship contribution statement

Baihui Huangfu: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. Wenjuan Cheng: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.ijcce.2024.10.002](https://doi.org/10.1016/j.ijcce.2024.10.002).

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